MIL-C-24231/5C(SHIPS)
30 August 1976
SUPERSEDING
MIL-C-24231/5B(SHIPS)
15 January 1970

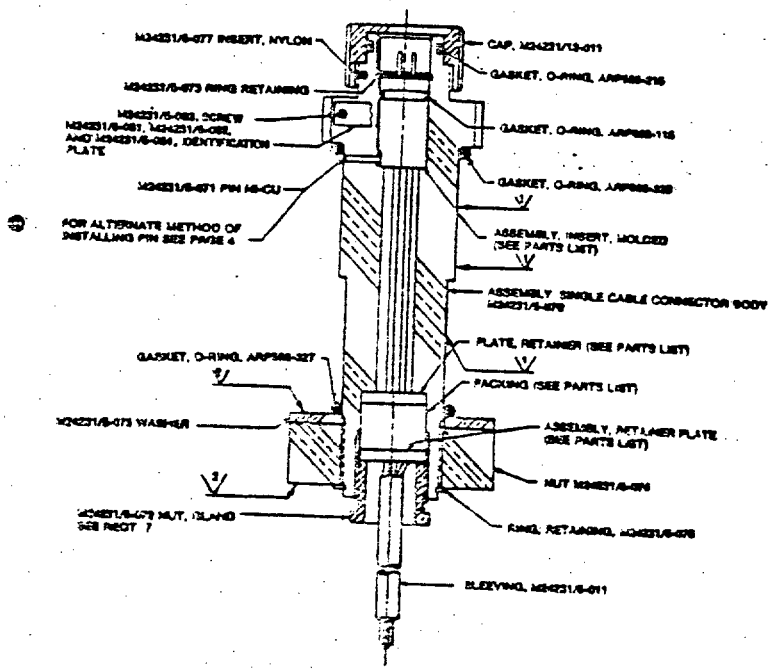
MILITARY SPECIFICATION SHEET

CONNECTORS, SINGLE CABLE, THREE-, FOUR- OR FIVE CONDUCTOR

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of MIL-C-24231(SHIPS)

ASSEMBLY, SINGLE CABLE CONNECTOR { 3 CONDUCTOR, M24231/5-001-SYM. NO. 517; 4 CONDUCTOR, M24231/5-002-SYM. NO. 517.11; 5 CONDUCTOR, M24231/5-003



ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ± MM INCH FOR FRACTIONS AND ± DECIMALS AND ANGULAR TOLERANCE IS ± DEGREE.

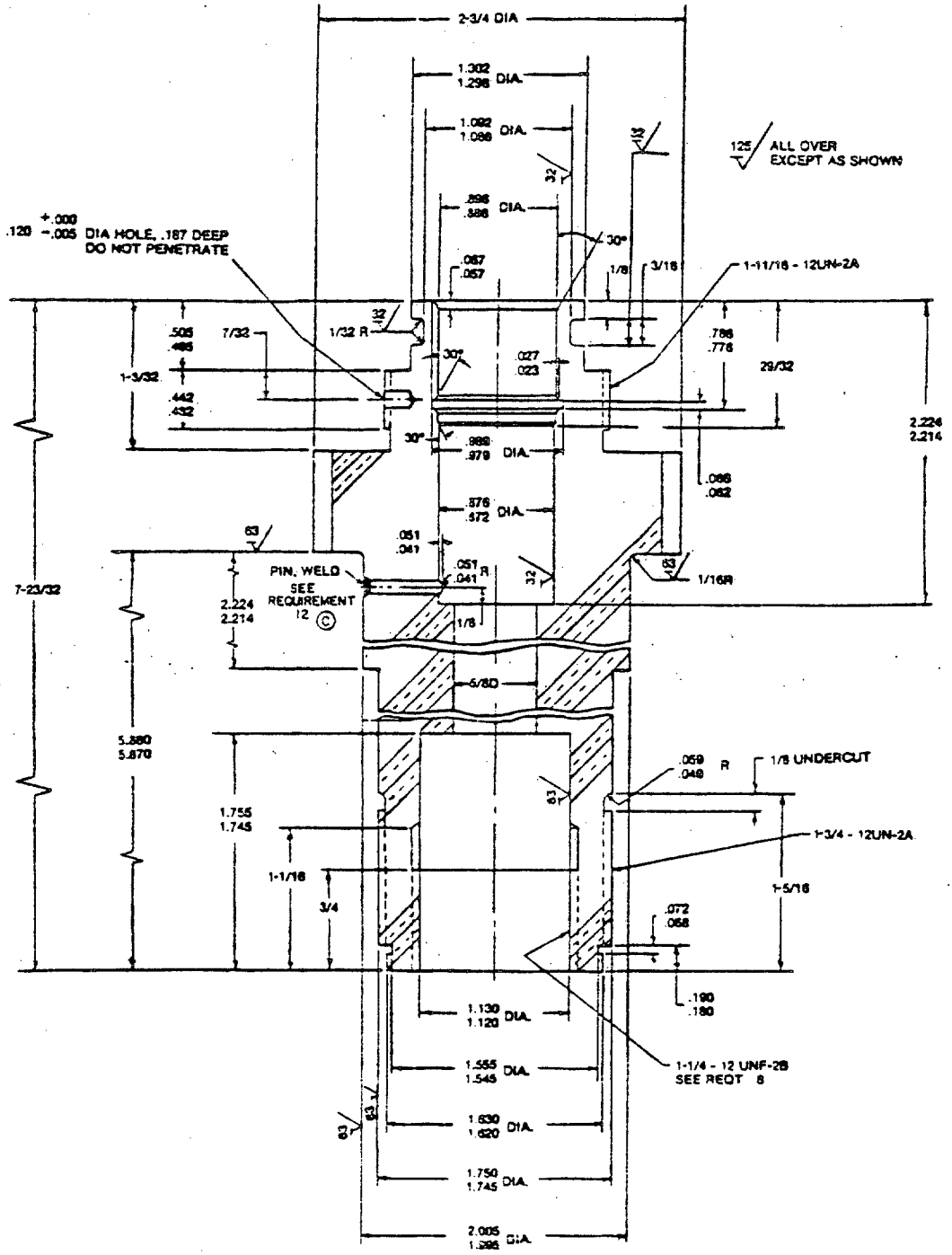
✓ BROWELL HARDNESS READING TO BE TAKEN AT THESE LOCATIONS

✓ BROWELL HARDNESS READING TO BE TAKEN AT THESE AREAS AT THE COUPON INTERVALS

© Denotes changes

FSC 5935

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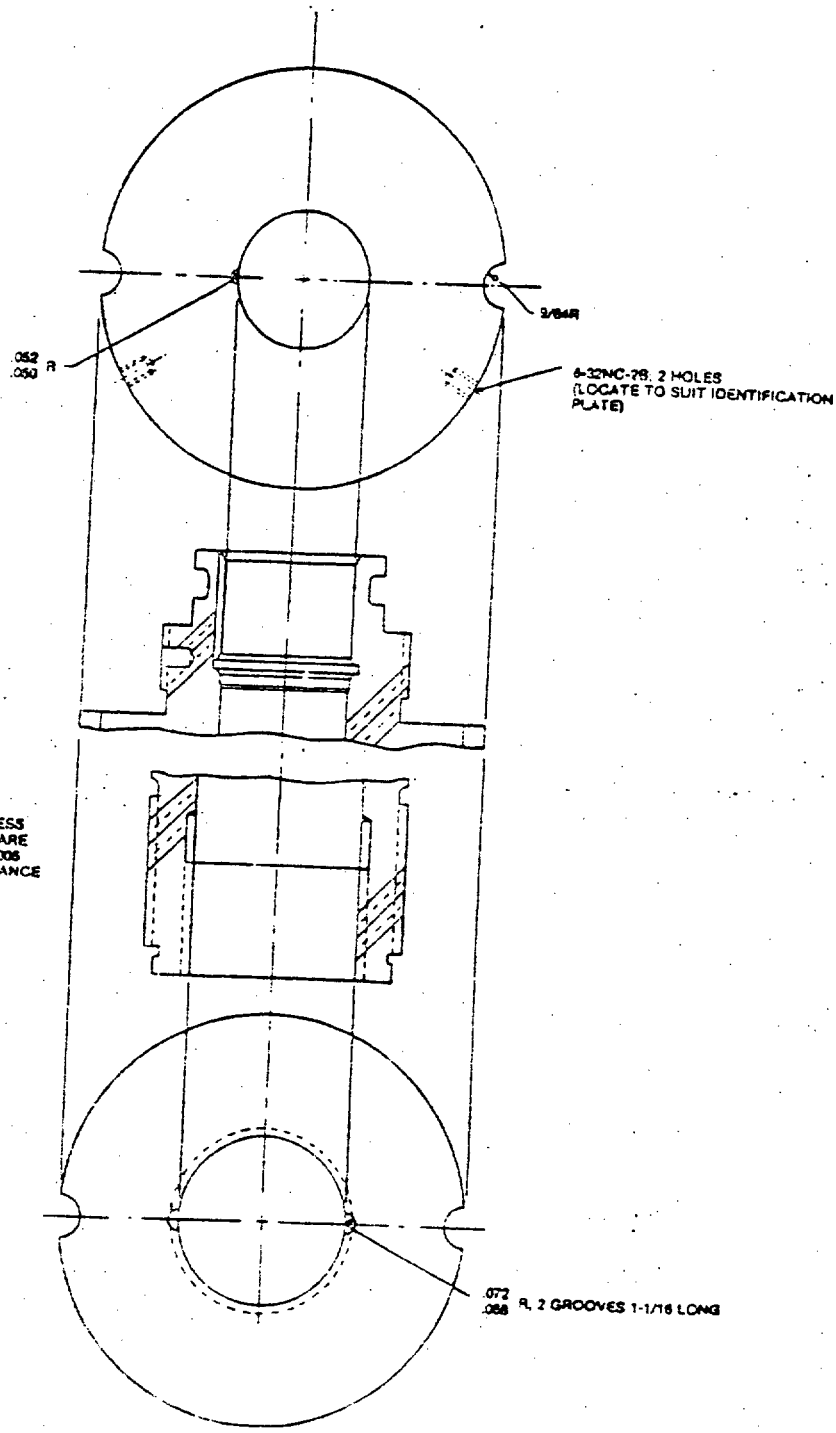


ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/64 INCH FOR FRACTIONS AND ±0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE IS ±1/2 DEGREE.

SH9872-1

ASSEMBLY, SINGLE CABLE CONNECTOR BODY
M24231/5-070

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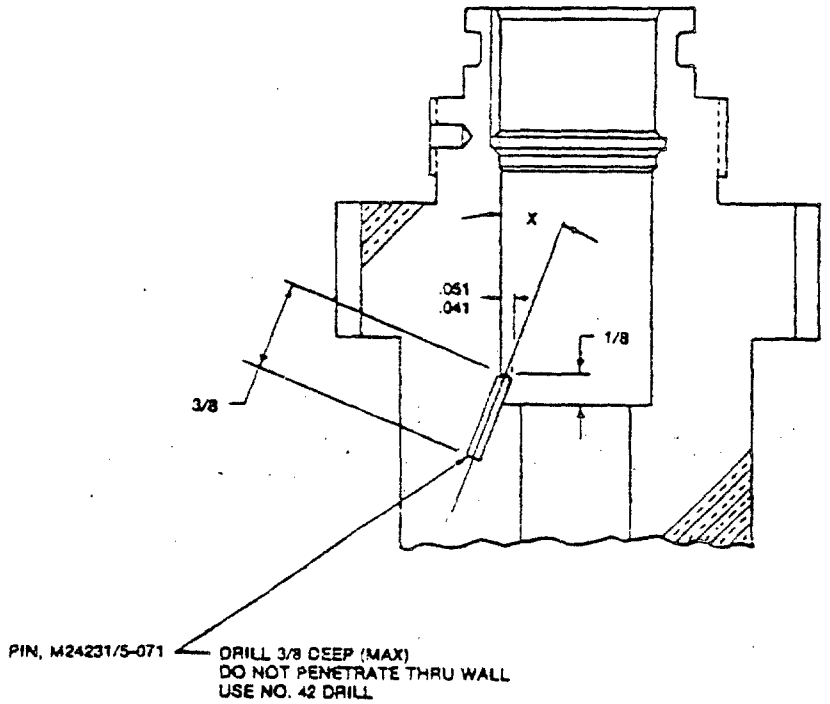
SH6872-2

ASSEMBLY SINGLE CABLE CONNECTOR BODY
M24231/5-070

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ALTERNATE METHOD
FOR
INSTALLATION OF GUIDE PIN
SEE REQUIREMENT 9

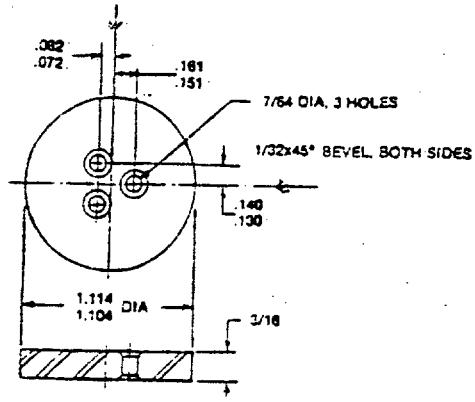


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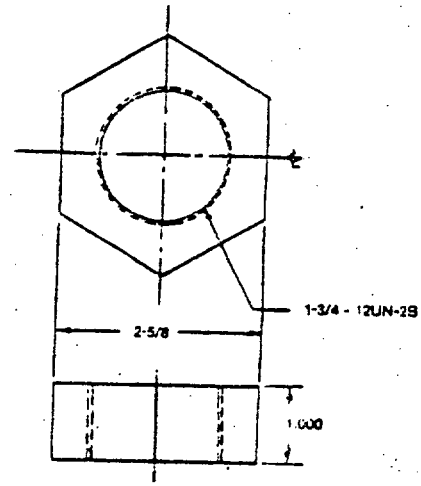
CONNECTOR BODY	"X" DEGREES
M24231/5-070	21-1/2
M24231/5-010	21-1/2
M24231/7-040	27
M24231/8-020	27
M24231/9-030	27
M24231/9-040	27
M24231/10-040	35
M24231/19-010	40

SH9672-3

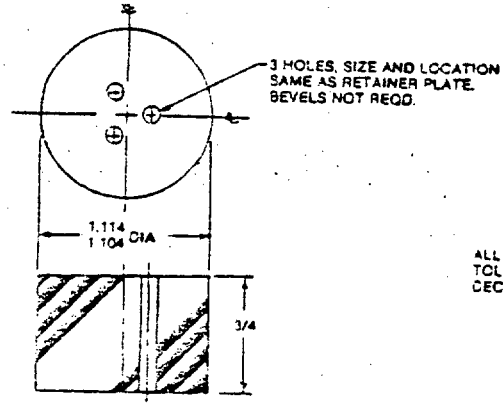
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PLATE, RETAINER
M24231/5-051

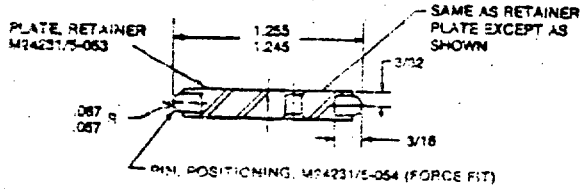


NUT
M24231/5-074

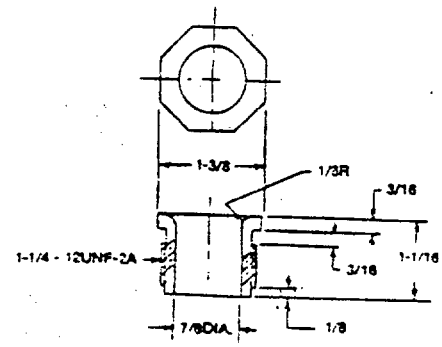


PACKING
M24231/5-052

ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/64 INCH FOR FRACTIONS AND ±0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE ±1/2 DEGREE.



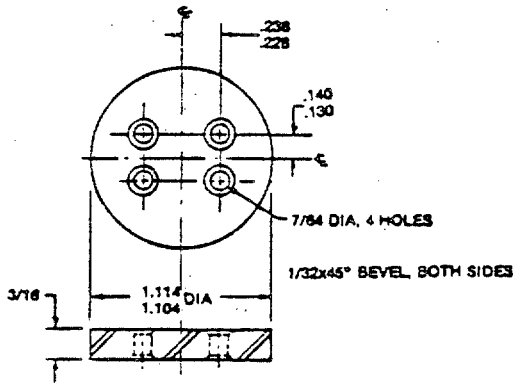
ASSEMBLY, RETAINER PLATE
M24231/5-050



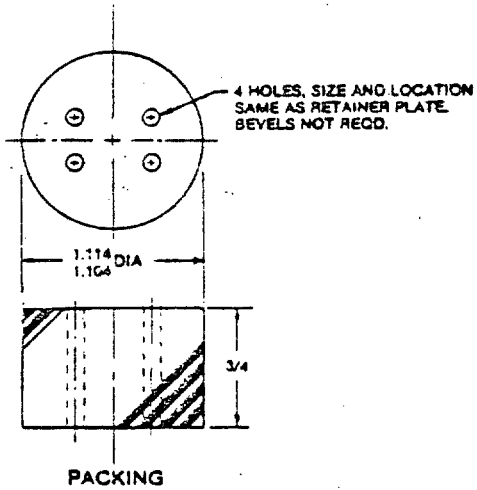
NUT, GLAND
M24231/5-072

S/0072-A

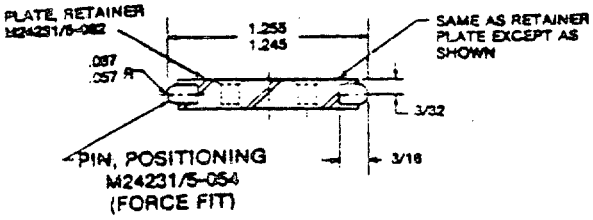
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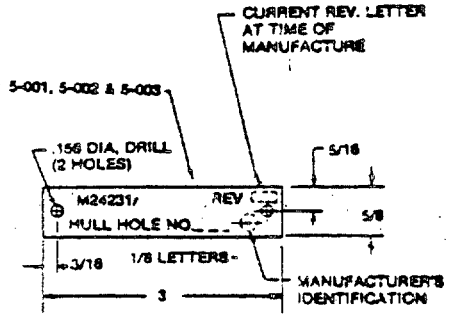
PLATE, RETAINER
M24231/5-081

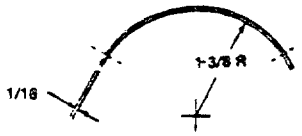


PACKING
M24231/5-063



ASSEMBLY, RETAINER PLATE
M24231/5-060





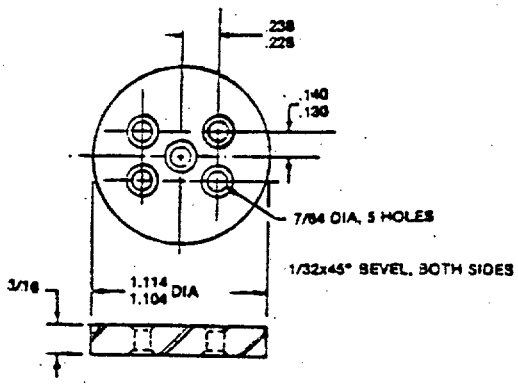
PLATE, IDENTIFICATION
M24231/5-081 (FOR M24231/5-001)
M24231/5-082 (FOR M24231/5-002)
M24231/5-084 (FOR M24231/5-003)

ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/64 INCH FOR FRACTIONS AND ±0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE IS ±1/2 DEGREE.

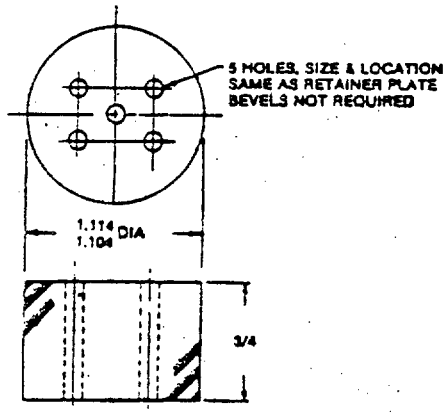
SH9672-5

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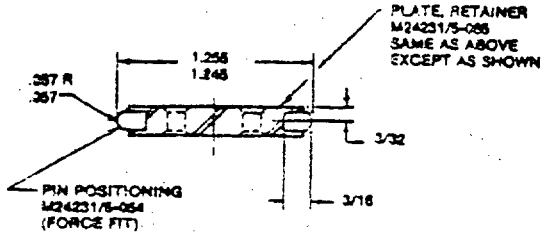
SEE REQUIREMENT 10



PLATE, RETAINER
M24231/5-064

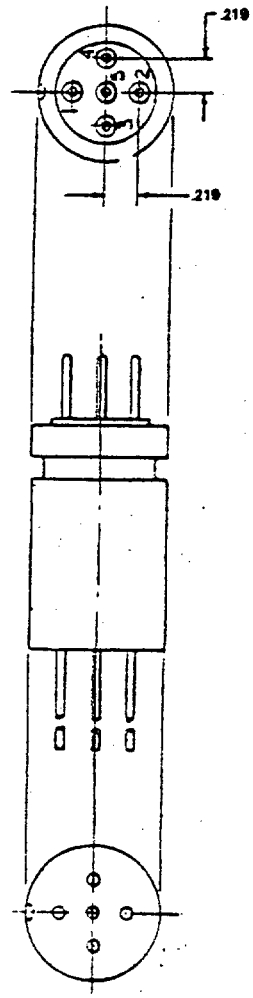


PACKING
M24231/5-085



ASSEMBLY, RETAINER PLATE
M24231/5-066

3H9672-6

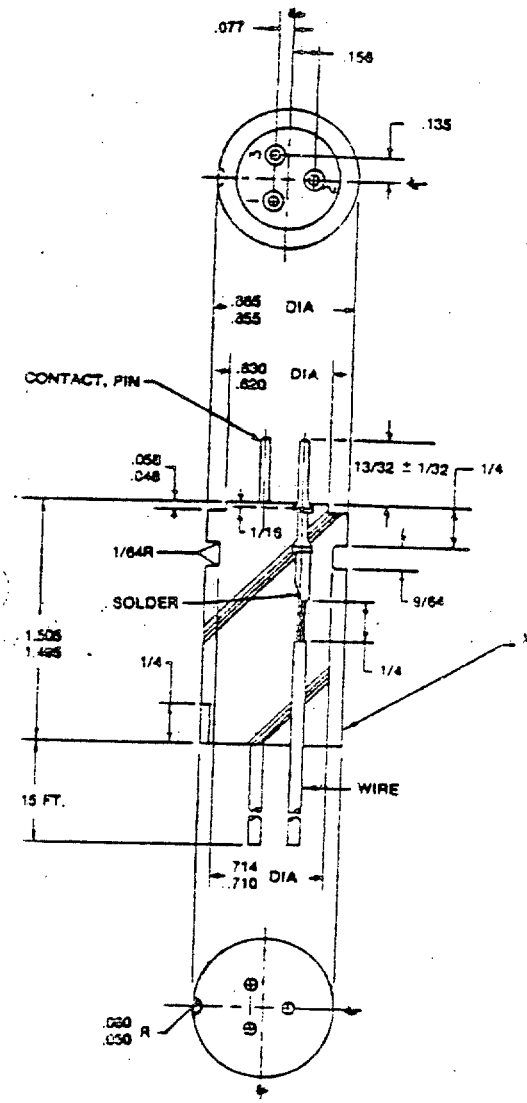


ASSY, INSERT, MOLDED (5 COND)
M24231/5-067
SAME AS M24231/5-010
EXCEPT AS SHOWN

ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/64 INCH FOR FRACTIONS AND ±0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE IS ±1/2 DEGREE.

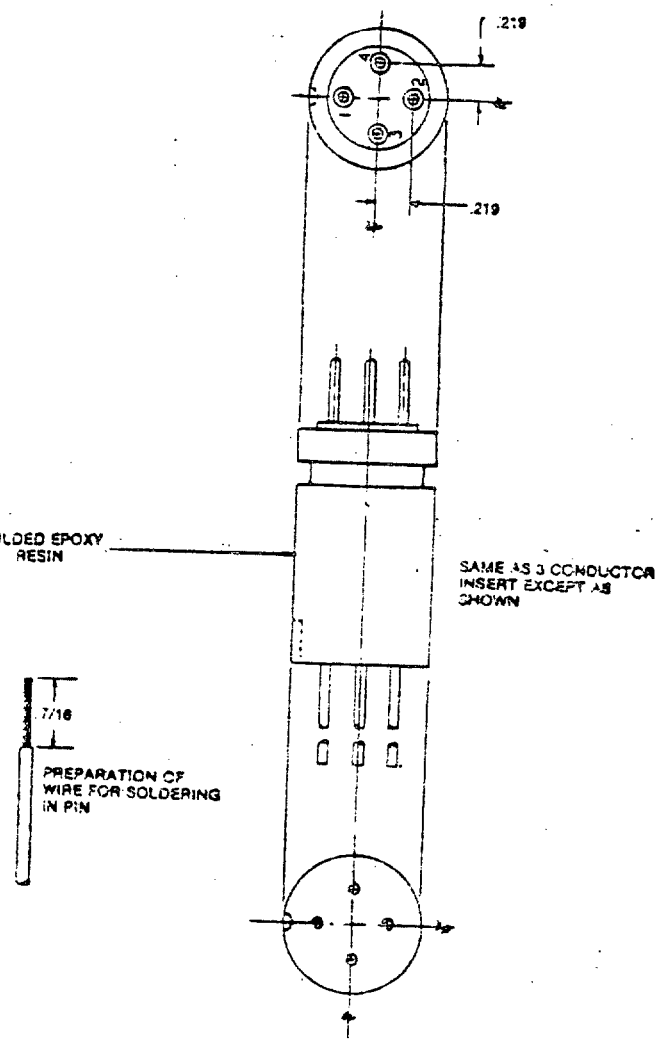
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ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±1/64 INCH FOR FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE IS ±1/2 DEGREE.



ASSEMBLY, INSERT, MOLDED (3 COND)
M24231/5-010

SH9672-7



ASSEMBLY, INSERT, MOLDED (4 COND)
M24231/5-020

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TABLE I - Parts list.

Part No.	Name	Quantity for		
		M24231/5-001	M24231/5-002	M24231/5-003
M24231/5-001	Assembly, Single Cable Connector (3-Conductor)	1		
M24231/5-002	Assembly, Single Cable Connector (4-Conductor)		1	
M24231/5-003	Assembly, Single Cable Connector (5-Conductor)			1
M24231/5-010	Assembly, Insert Molded (3-Conductor)	1		
M24231/5-011	Sleeving (0.3601 D.X. 020 THK)	14 ft.	14 ft.	14 ft.
M24231/5-020	Assembly, Insert Molded (4-Conductor)		1	
M24231/5-050	Assembly, Retainer Plate	1		
M24231/5-051	Plate, Retainer	1		
M24231/5-052	Packing	1		
M24231/5-053	Plate, Retainer	1		
M24231/5-054	Pin, Positioning	2		
M24231/5-060	Assembly, Retainer Plate		2	2
M24231/5-061	Plate, Retainer		1	
M24231/5-062	Plate, Retainer		1	
M24231/5-063	Packing		1	
M24231/5-064	Plate, Retainer		1	
M24231/5-065	Plate, Retainer			1
M24231/5-066	Assembly, Retainer Plate			1
M24231/5-067	Assembly, Insert Molded (5-Conductor)			1
M24231/5-070	Assembly, Single Cable Connector Body			1
M24231/5-071	Pin (0.092D) Length to suit	1	1	1
M24231/5-072	Nut, Bland	1	1	1
M24231/5-073	Ring, Retaining	1	1	1
M24231/5-074	Nut, Nickel Copper Aluminum Alloy	1	1	1
M24231/5-075	Washer, Nickel Copper Aluminum Alloy	1	1	1
M24231/5-076	Ring, Retaining	1	1	1
M24231/5-077	Insert, Nylon (0.125 ± 0.002D)	0.209 long	0.209 long	0.209 long
M24231/5-081	Plate, Identification	1		
M24231/5-082	Plate, Identification		1	
M24231/5-083	Screw (6-32 X 3/16 CRES)	2	2	2
M24231/5-084	Plate, Identification			1
M24231/5-085	Packing			1
ARP568-115	O-Ring Gasket	1	1	1
ARP568-215	O-Ring Gasket	1	1	1
ARP568-327	O-Ring Gasket	1	1	1
ARP568-329	O-Ring Gasket	1	1	1
M24231/12-011	Cap	1	1	1

REQUIREMENTS:

1. Dimensions and configuration: See figures.

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REQUIREMENTS: (Conc'd)

2. Material for pin M24231/5-071 shall be nickel-copper alloy in accordance with Class A of QQ-N-281.
3. Positioning pin M24231/5-054 shall be nickel-copper alloy in accordance with Class A of QQ-N-281.
4. Retaining ring M24231/5-073 shall be Spiralox, PRD-99-M, or equal.
5. Retaining ring M24231/5-076 shall be Truarc, 5100-162-W, or equal.
6. Dimensions for washer M24231/5-075 shall be 3-1/2 inches O.D. by 1.760 inches I.D. by 0.162 inch thick.
7. The gland nut shall be tightened until the packing tightens against wires. This can be determined by moving wires gently back and forth until wire is seized by packing. The gland nut shall be turned approximately one-fourth of a turn more to hold the wires in place. DO NOT OVERTIGHTEN the gland nut or attempt to tighten down metal to metal, as this may damage the wires.
8. Full or partial threads shall not extend more than 0.083 inch beyond the 3/4-inch thread length.
9. (C) This alternate method of installing the aligning pin eliminates the need for welding in the fabrication of the single cable connector body. This gives the manufacturing activity the option to make the body, nut and washer from any age hardened ni-cu-al alloy in accordance with class A of QQ-N-286 that will meet the mechanical properties listed in paragraph 3.3.2 of MIL-C-24231. In that case, the chemical and physical certification obtained from the supplier of the ni-cu-al alloy will satisfy certification requirements when supplemented by the required traceability markings identifying the finished product to the bar stock on which the chemical and physical analysis was performed. Install pin with locking and retaining compound, MIL-S-22473, Grade A. See page 4.
10. (C) Other special 5-conductor arrangements; i.e., 2SWU-1 and B-16 wire combinations shall be the responsibility of the installing activity.
11. (C) The retainer plates and retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with class A, form 1, hot finished in accordance with QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with class A, form 1, cold drawn, in accordance with QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.
12. (C) Welding of the pin (/5-071) is allowed after the hull fitting body has been age-hardened to achieve its required physical properties. This gives the manufacturing activity the option to make the body, nut and washer from any age hardened ni-cu-al alloy in accordance with class A of QQ-N-286 that will meet the mechanical properties listed in paragraph 3.3.2 of MIL-C-24231. In that case, the chemical and physical certification obtained from the supplier of the ni-cu-al alloy will satisfy certification requirements when supplemented by the required traceability markings identifying the finished product to the bar stock on which the chemical and physical analysis was performed.

Preparing activity:
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(Project 5935-N842-5)

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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